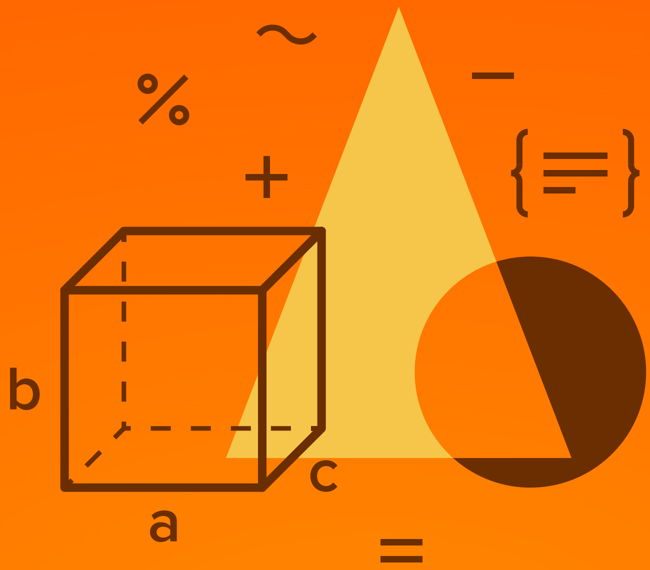
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success



MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Probability

English Medium

EXERCISE (O-1)

Straight Objective Type

1. 6 married couples are standing in a room. If 4 people are chosen at random, then the chance that exactly one married couple is among the 4 is-

(A) $\frac{16}{33}$ (B) $\frac{8}{33}$ (C) $\frac{17}{33}$ (D) $\frac{24}{33}$

PB0001

2. The probability that a positive two digit number selected at random has its tens digit at least three more than its unit digit is -

(A) 14/45 (B) 7/45 (C) 36/45 (D) 1/6

PB0002

3. A 5 digit number is formed by using the digits 0,1,2,3,4 & 5 without repetition. The probability that the number is divisible by 6 is :

(A) 8% (B) 17% (C) 18% (D) 36%

PB0003

4. Mr. A forgot to write down a very important phone number. All he remembers is that it started with 713 and that the next set of 4 digit involved are 1,7 and 9 with one of these numbers appearing twice. He guesses a phone number and dials randomly. The odds in favour of dialing the correct telephone number, is -

(A) 1 : 35 (B) 1 : 71 (C) 1 : 23 (D) 1 : 36

PB0010

5. There are 4 defective items in a lot consisting of 10 items. From this lot we select 5 items at random. The probability that there will be 2 defective items among them is -

(A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{5}{21}$ (D) $\frac{10}{21}$

PB0017

6. If two of the 64 squares are chosen at random on a chess board, the probability that they have a side in common is -

(A) 1/9 (B) 1/18 (C) 2/7 (D) none

PB0019

7. Two red counters, three green counters and 4 blue counters are placed in a row in random order. The probability that no two blue counters are adjacent is -

(A) $\frac{7}{99}$ (B) $\frac{7}{198}$ (C) $\frac{5}{42}$ (D) none

PB0020

8. South African cricket captain lost the toss of a coin 13 times out of 14. The chance of this happening was

(A) $\frac{7}{2^{13}}$ (B) $\frac{1}{2^{13}}$ (C) $\frac{13}{2^{14}}$ (D) $\frac{13}{2^{13}}$

PB0021

9. There are ten prizes, five A's, three B's and two C's, placed in identical sealed envelopes for the top ten contestants in a mathematics contest. The prizes are awarded by allowing winners to select an envelope at random from those remaining. When the 8^{th} contestant goes to select the prize, the probability that the remaining three prizes are one A, one B and one C, is

(A) 1/4 (B) 1/3 (C) 1/12 (D) 1/10

PB0022

10. A coin is tossed and a die is thrown, then the probability that the outcome will be a head or a number greater than 4.

PB0023

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

11. A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant chosen has the value non negative is :

(A) 3/16 (B) 6/16 (C) 10/16 (D) 13/16

PB0037

12. A card is drawn & replaced in an ordinary pack of 52 playing cards. Minimum number of times must a card be drawn so that there is atleast an even chance of drawing a heart, is

(A) 2 (B) 3 (C) 4 (D) more than four

PB0038

13. A license plate is 3 capital letters (of English alphabets) followed by 3 digits. If all possible license plates are equally likely, the probability that a plate has either a letter palindrome or a digit palindrome (or both), is

(A) $\frac{7}{52}$ (B) $\frac{9}{65}$ (C) $\frac{8}{65}$ (D) none

PB0039

14. Whenever horses a, b, c race together, their respective probabilities of winning the race are 0.3, 0.5 and 0.2 respectively. If they race three times the probability that "the same horse wins all the three races" and the probability that a, b, c each wins one race, are respectively (Assume no dead heat)

(A) $\frac{8}{50};\frac{9}{50}$ (B) $\frac{16}{100},\frac{3}{100}$ (C) $\frac{12}{50};\frac{15}{50}$ (D) $\frac{10}{50};\frac{8}{50}$

PB0040

15. Two cubes have their faces painted either red or blue. The first cube has five red faces and one blue face. When the two cubes are rolled simultaneously, the probability that the two top faces show the same colour is 1/2. Number of red faces on the second cube, is

(A) 1 (B) 2 (C) 3 (D) 4

PB0041

16. A committee of three persons is to be randomly selected from a group of three men and two women and the chair person will be randomly selected from the committee. The probability that the committee will have exactly two women and one man, and that the chair person will be a woman, is/are

(A) 1/5 (B) 8/15 (C) 2/3 (D) 3/10

PB0042

17. A bag contains 3 R & 3 G balls and a person draws out 3 at random. He then drops 3 blue balls into the bag & again draws out 3 at random. The chance that the 3 later balls being all of different colours is

(A) 15% (B) 20% (C) 27% (D) 40%

PB0052

18. An urn contains 10 balls coloured either black or red. When selecting two balls from the urn at random, the probability that a ball of each colour is selected is $8/15$. Assuming that the urn contains more black balls than red balls, the probability that at least one black ball is selected, when selecting two balls, is

(A) $\frac{18}{45}$ (B) $\frac{30}{45}$ (C) $\frac{39}{45}$ (D) $\frac{41}{45}$

PB0056

19. The number 'a' is randomly selected from the set {0, 1, 2, 3, ... 98, 99}. The number 'b' is selected from the same set. Probability that the number $3^a + 7^b$ has a digit equal to 8 at the units place, is

(A) $\frac{1}{16}$ (B) $\frac{2}{16}$ (C) $\frac{4}{16}$ (D) $\frac{3}{16}$

PB0060

20. If a, b and c are three numbers (not necessarily different) chosen randomly and with replacement from the set {1, 2, 3, 4, 5}, the probability that (ab + c) is even, is

(A) $\frac{35}{125}$ (B) $\frac{59}{125}$ (C) $\frac{64}{125}$ (D) $\frac{75}{125}$

PB0091

21. An ant is situated at the vertex A of the triangle ABC. Every movement of the ant consists of moving to one of other two adjacent vertices from the vertex where it is situated. The probability of going to any of the other two adjacent vertices of the triangle is equal. The probability that at the end of the fourth movement the ant will be back to the vertex A, is :

(A) 4/16 (B) 6/16 (C) 7/16 (D) 8/16

PB0062

22. Mr. A and Mr. B each have a bag that contains one ball of each of the colours blue, green, orange, red and violet. 'A' randomly selects one ball from his bag and puts it into B's bag. 'B' then randomly selects one ball from his bag and puts it into A's bag. The probability that after this process the contents of the two bags are the same, is

(A) 1/6 (B) 1/5 (C) 1/3 (D) 1/2

PB0065

23. A box contains 10 tickets numbered from 1 to 10. Two tickets are drawn one by one without replacement. The probability that the "absolute value of difference between the first drawn ticket number and the second is not less than 4" is

(A) $\frac{7}{30}$ (B) $\frac{7}{15}$ (C) $\frac{11}{30}$ (D) $\frac{10}{30}$

PB0067

24. The probability that a radar will detect an object in one cycle is p. The probability that the object will be detected in n cycles is :

(A) $1 - p^n$ (B) $1 - (1 - p)^n$ (C) p^n (D) $p(1 - p)^{n-1}$

PB0075

25. Three numbers are chosen at random without replacement from $\{1, 2, 3, \ldots\ldots, 10\}$. The probability that the minimum of the chosen numbers is 3 or their maximum is 7 is

(A) 1/2 (B) 1/3 (C) 1/4 (D) 11/40

PB0077

26. In a maths paper there are 3 sections A, B & C. Section A is compulsory. Out of sections B & C a student has to attempt any one. Passing in the paper means passing in A & passing in B or C. The probability of the student passing in A, B & C are p, q & 1/2 respectively student select randmoly one of the option out of B & C. If the probability that the student is successful is 1/2 then possible choosen can be :

(A) $p = q = 1$ (B) $p = q = 1/2$

(C) $p = 1, q = 0$ (D) $p = 1, q = 1/2$

PB0085

27. If C and D are two events such that $C \subset D$ and $P(D) \neq 0$, then the correct statement among the following is :-

(1) $P(C|D) < P(C)$ (2) $P(C|D) = \frac{P(D)}{P(C)}$

(3) $P(C|D) = P(C)$ (4) $P(C|D) \geq P(C)$

PB0155

28. Consider 5 independent Bernoulli's trials each with probability of success p. If the probability of at least one failure is greater than or equal to $\frac{31}{32}$, then p lies in the interval :-

(1) $\left[0,\frac{1}{2}\right]$ (2) $\left(\frac{11}{12},1\right]$ (3) $\left(\frac{1}{2},\frac{3}{4}\right]$ (4) $\left(\frac{3}{4},\frac{11}{12}\right]$

PB0156

29. Let A, B, C be pairwise independent events with $P(C) > 0$ and $P(A \cap B \cap C) = 0$. Then $P(A^c \cap B^c | C)$ is equal to :

(1) $P(A^c) - P(B)$ (2) $P(A) - P(B^c)$ (3) $P(A^c) + P(B^c)$ (4) $P(A^c) - P(B^c)$

PB0157

30. Three numbers are chosen at random without replacement from $\{1, 2, 3, ... , 8\}$. The probability that their minimum is 3, given that their maximum is 6, is :

(1) $\frac{2}{5}$ (2) $\frac{3}{8}$ (3) $\frac{1}{5}$ (4) $\frac{1}{4}$

PB0158

EXERCISE (O-2)

Multiple Correct Answer Type

1. A boy has a collection of blue and green marbles. The number of blue marbles belong to the sets {2, 3, 4, ... 13}. If two marbles are chosen simultaneously and at random from his collection, then the probability that they have different colour is $1/2$. Possible number of blue marbles is :

(A) 2 (B) 3 (C) 6 (D) 10

PB0099

2. If A & B are two events such that $P(B) \neq 1$, B^C denotes the event complementary to B, then

(A) $P\left(A/B^C\right) = \dfrac{P(A) - P(A \cap B)}{1 - P(B)}$

(B) $P(A \cap B) \geq P(A) + P(B) - 1$

(C) $P(A) >< P\left(A/B\right)$ according as $P\left(A/B^C\right) >< P(A)$

(D) $P\left(A/B^C\right) + P\left(A^C/B^C\right) = 1$

PB0100

3. For $P(A) = \dfrac{3}{8}$; $P(B) = \dfrac{1}{2}$; $P(A \cup B) = \dfrac{5}{8}$ which of the following do/does hold good?

(A) $P\left(A^c/B\right) = 2P\left(A/B^c\right)$ (B) $P(B) = P\left(A/B\right)$

(C) $15P\left(A^c/B^c\right) = 8P\left(B/A^c\right)$ (D) $P\left(A/B^c\right) = P(A \cap B)$

PB0101

4. If E_1 and E_2 are two events such that $P(E_1) = 1/4$, $P(E_2/E_1) = 1/2$ and $P(E_1/E_2) = 1/4$

(A) then E_1 and E_2 are independent

(B) E_1 and E_2 are exhaustive

(C) E_2 is twice as likely to occur as E_1

(D) Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in G.P.

PB0102

5. Two events A and B are such that the probability that at least one of them occurs is 5/6 and both of them occurring simultaneously is 1/3. If the probability of not occurrence of B is 1/2 then

(A) A and B are equally likely (B) A and B are independent

(C) $P(A/B) = 2/3$ (D) $3\,P(A) = 4\,P(B)$

PB0103

6. The probabilities of events, $A \cap B$, A, B & $A \cup B$ are respectively in A.P. with probability of second term equal to the common difference. Therefore the events A and B are

(A) mutually exclusive (B) independent

(C) such that one of them must occur (D) such that one is twice as likely as the other

PB0104

7. A box contains 11 tickets numbered from 1 to 11. Six tickets are drawn simultaneously at random. Let E_1 denotes the event that the sum of the numbers on the tickets drawn is even and E_2 denotes the event that the sum of the numbers on the tickets drawn is odd. Which of the following hold good?

(A) E_1 and E_2 are equally likely
(B) E_1 and E_2 are exhaustive
(C) $P(E_2) > P(E_1)$
(D) $P(E_1/E_2) = P(E_2 / E_1)$

PB0105

8. If $\overline{E}$ & $\overline{F}$ are the complementary events of events E & F respectively & if $0 < P(F) < 1$, then :

(A) $P(E|F) + P(\overline{E}|F) = 1$
(B) $P(E|F) + P(E|\overline{F}) = 1$
(C) $P(\overline{E}|F) + P(E|\overline{F}) = 1$
(D) $P(E|\overline{F}) + P(\overline{E}|\overline{F}) = 1$

PB0106

9. Probability of **n** heads in **2n** tosses of a fair coin can be given by

(A) $\prod_{r=1}^{n}\left(\frac{2r-1}{2r}\right)$
(B) $\prod_{r=1}^{n}\left(\frac{n+r}{2r}\right)$
(C) $\sum_{r=0}^{n}\left(\frac{{}^{n}C_r}{2^n}\right)^2$
(D) $\frac{\sum_{r=0}^{n}\left({}^{n}C_r\right)^2}{\left(\sum_{r=0}^{n}{}^{n}C_r\right)^2}$

PB0107

10. Which of the following statements is/are True?

(A) A fair coin is tossed n times where n is a positive integer. The probability that n^{th} toss results in head is 1/2.

(B) The conditional probability that the n^{th} toss results in head given that first $(n-1)$ tosses results in head is $1/2^n$

(C) Let E and F be the events such that F is neither impossible nor sure. If $P(E/F) > P(E)$ then $P(E/F^c) > P(E)$

(D) If A, B and C are independent then the events $(A \cup B)$ and C are independent.

PB0108

11. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If P(T) denotes the probability of occurrence of the event T, then -

(A) $P(E) = \frac{4}{5}$, $P(F) = \frac{3}{5}$
(B) $P(E) = \frac{1}{5}, P(F) = \frac{2}{5}$
(C) $P(E) = \frac{2}{5}, P(F) = \frac{1}{5}$
(D) $P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$

PB0182

12. A ship is fitted with three engines E_1, E_2 and E_3. The engines function independently of each other with respective probabilities $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{4}$. For the ship to be operational at least two of its engines must function. Let X denote the event that the ship is operational and X_1, X_2, X_3 denotes respectively the events that the engines E_1, E_2 and E_3 are functioning. Which of the following is (are) true?

(A) $P\left[X_1^c \mid X\right] = \frac{3}{16}$

(B) P[Exactly two engines of ship are functioning $|X] = \frac{7}{8}$

(C) $P[X \mid X_2] = \frac{5}{16}$

(D) $P[X \mid X_1] = \frac{7}{16}$

PB0183

13. Let X and Y be two events such that $P(X \mid Y) = \frac{1}{2}, P(Y \mid X) = \frac{1}{3}$ and $P(X \cap Y) = \frac{1}{6}$. Which of the following is(are) correct ?

(A) $P(X \cup Y) = \frac{2}{3}$

(B) X and Y are independent

(C) X and Y are not independent

(D) $P(X^C \cap Y) = \frac{1}{3}$

PB0185

14. A lot contains 50 defective & 50 non defective bulbs. Two bulbs are drawn at random, one at a time, with replacement. The events A, B, C are defined as :

A = { the first bulb is defective};

B = { the second bulb is non defective}

C = { the two bulbs are both defective or both non defective}. Then which of the following is/are correct.

(A) A,B,C are pair wise independent

(B) A,B,C are independent

(C) A,B,C are not pair wise independent

(D) A,B,C are not independent

PB0124

15. A bomber wants to destroy a bridge . Two bombs are sufficient to destroy it . If four bombs are dropped, if the chance of a bomb hitting the target is 0.4. Then the probability that it is destroyed, is less than

PB0126

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{3}{4}$

(D) $\frac{4}{5}$

16. Before a race the chance of three runners, A, B, C were estimated to be proportional to 5, 3, 2, but during the race A meets with an accident which reduces his chance to 1/3. Now P(B) and P(C) denotes the respective probabilities for winning B & C, then

(A) $P(B)=\frac{2}{5}$ (B) $P(C)=\frac{4}{15}$ (C) $P(B)=\frac{1}{4}$ (D) $P(C)=\frac{1}{6}$

PB0136

17. A box contains 11 tickets numbered from 1 to 11. Two tickets are drawn similtaneously at random. Let E_1 denotes the event that the sum of the numbers on the tickets drawn is even and E_2 denotes the event that the sum of the numbers on the tickets drawn is odd

Which of the following hold good ? **PB0214**

(A) $P(E_1/E_2) = P(E_2/E_1)$ (B) E_1 and E_2 are exhaustive

(C) $P(E_2) > P(E_1)$ (D) E_1 and E_2 are equally likely

18. In an experimental performance of a single throw of a pair of unbiased normal dice, three events E_1, E_2 & E_3 are defined as follows : **PB0215**

E_1 : getting a prime numbered face on each dice

E_2 : getting the same number on each dice

E_3 : getting a sum total of dots on two dice equal to 8. Then :

(A) the events E_1, E_2 & E_3 are mutually exclusive

(B) the events E_1, E_2 & E_3 are not pairwise mutually exclusive

(C) the events E_1, E_2 are independent

(D) $P(E_3 \mid E_1) = 2/9$.

19. The probability that a bulb produced by a factory will fuse after an year of use is 0.1. Then the probability that out of 4 such bulbs **PB0216**

(A) None of then bulb will fuse after an year of use is $\frac{9^4}{10^4}$

(B) More then three bulbs will fuse after an year of use is $\frac{1}{10^4}$

(C) Not more then three bulbs will fuse after an year of use is $\frac{9999}{10000}$

(D) All the bulbs will fuse after an year of use is $\frac{1}{10^4}$

20. An unbiased coin is tossed n times. Let X denote the number of times head occurs. If P(X = 4), P(X = 5) and P(X = 6) are in AP, then the value of n can be **PB0217**

(A) 7 (B) 10 (C) 12 (D) 14

21. Ram post a letter to Ghanshyam. It is known that one letter out of 10 letters do not reach its destination. If it is certain that Ghanshyam will reply if he receives the letter. If A denotes the event that Ghanshyam receives the letter and B that Ram gets a reply, then - **PB0218**

(A) $P(B) = \frac{81}{100}$ (B) $P(A \cap B) = \frac{81}{100}$

(C) $P\left(\frac{A}{\overline{B}}\right) = \frac{9}{19}$ (D) $P(A \cup B) = \frac{9}{10}$

22. Before a race the chance of three runners A, B & C were estimated to be proportional to 5, 3 & 2 respectively but during the race A meets with an accident which reduces his chance to 1/3. If the respective chances of B and C are P(B) and P(C) then - **PB0219**

(A) $P(B) = \frac{2}{5}$ (B) $P(C) = \frac{4}{15}$ (C) $P(C) = \frac{2}{5}$ (D) $P(B) = \frac{4}{15}$

23. Two real numbers, x & y are selected at random. Given that $0 \le x \le 1;\ 0 \le y \le 1$. Let A be the event that $y^2 \le x$; B be the event that $x^2 \le y$, then - **PB0220**

(A) $P(A \cap B) = \frac{1}{3}$ (B) A & B are exhaustive events

(C) A & B are mutually exclusive (D) A & B are independent events.

24. Three dice are rolled simultaneously, then the chance that exactly two of the dice show the same number is more than - **PB0221**

(A) $\frac{5}{18}$ (B) $\frac{5}{9}$ (C) $\frac{5}{13}$ (D) $\frac{1}{3}$

25. Suppose that M is a natural number with the property that if x is chosen randomly from the set $\{1, 2, \ldots\ldots, 1000\}$ the probability that x is divisor of M is $\frac{1}{2}$. If $M \le 1000$ then maximum possible value of M is divisible by - **PB0222**

(A) 2 (B) 61 (C) 3 (D) 5

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for question nos. 1 and 2

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with a 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively

1. The chance she will be successful, is

(A) 0.28 (B) 0.38 (C) 0.48 (D) 0.58

PB0072

2. Given that she is successful, the chance she studied for 4 hours, is

(A) $\frac{6}{12}$ (B) $\frac{7}{12}$ (C) $\frac{8}{12}$ (D) $\frac{9}{12}$

PB0072

Paragraph for question nos. 3 and 4

Two buses A and B are scheduled to arrive at a town central bus station at noon. The probability that bus A will be late is 1/5. The probability that bus B will be late is 7/25. The probability that the bus B is late given that bus A is late is 9/10. Then the probabilities

3. The probability that neither bus will be late on a particular day is

(A) $\frac{2}{25}$ (B) $\frac{18}{25}$ (C) $\frac{7}{10}$ (D) $\frac{12}{25}$

4. The probability that bus A is late given that bus B is late, is

(A) $\frac{12}{28}$ (B) $\frac{22}{28}$ (C) $\frac{18}{28}$ (D) $\frac{2}{28}$

PB0078

Paragraph for question 5 and 6

Two whole numbers are randomly selected and multiplied. Consider two events E_1 and E_2 defined as

E_1 : Their product is divisible by 5

E_2 : Unit's place in their product is 5.

5. Probability $P(E_1)$ is

6. Probability $P\left(\frac{E_2}{E_1}\right)$ is

PB0098

Paragraph for question nos. 7 and 8

A uniform unbiased die is constructed in the shape of a regular tetrahedron with faces numbered 2, 2, 3 and 4 and the score is taken from the face on which the die lands. If two such dice are thrown together.

7. Probability of scoring exactly 6 on each of 3 successive throws, can not be

(A) $\left(\frac{2}{16}\right)^3$ (B) $\left(\frac{3}{16}\right)^3$ (C) $\left(\frac{5}{16}\right)^3$ (D) $\left(\frac{6}{16}\right)^3$

8. Probability of scoring more than 4 on at least one of the three successive throws, is greater than

(A) $\frac{63}{64}$ (B) $\frac{21}{64}$ (C) $\frac{42}{64}$ (D) $\frac{3}{64}$

PB0132

9.

	List-I		List-II
(I)	If the probability that units digit in square of an even integer is 4 is p, then the value of 5p is	(P)	1
(II)	If A and B are independent events and $P(A \cap B) = \frac{1}{6}$, $P(A) = \frac{1}{3}$, then $6P\left(\frac{B}{\overline{A}}\right) =$	(Q)	2
(III)	One mapping is selected at random from all mappings of the set S = {1, 2, 3, ..., n} into itself. If the probability that the mapping is one-one is $\frac{3}{32}$, then the value of n is	(R)	3
(IV)	A boy has 20% chance of hitting at a target. Let p denote the probability of hitting the target for the first time at the n^{th} trial. If p satisfies the inequality $625p^2 - 175p + 12 < 0$, then value of n is	(S)	4

(A) (I) → (Q), (II) → (R), (III) → (S), (IV) → (R)

(B) (I) → (R), (II) → (Q), (III) → (R), (IV) → (S)

(C) (I) → (P), (II) → (S), (III) → (R), (IV) → (Q)

(D) (I) → (Q), (II) → (P), (III) → (Q), (IV) → (R)

PB0223

10. **Column-I** **Column-II**

	Column-I		Column-II
(A)	Two different numbers are taken from the set {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. The probability that their sum and positive difference, are both multiple of 4, is $x/55$ then x equals	(P)	4
		(Q)	6
(B)	There are two red, two blue, two white and certain number (greater than 0) of green socks in a drawer. If two socks are taken at random from the drawer without replacement, the probability that they are of the same colour is 1/5 then the number of green socks are	(R)	8
(C)	A drawer contains a mixture of red socks and blue socks, atmost 17 in all. It so happens that when two socks are selected randomly without replacement, there is a probability of exactly $1/2$ that both are red or both are blue. The largest possible number of red socks in the drawer that is consistent with this data, is	(S)	10

PB0109

EXERCISE (O-4)

Numerical Grid Type

1. There are 2 groups of subjects one of which consists of 5 science subjects & 3 engg. subjects & other consists of 3 science & 5 engg. subjects. An unbiased die is cast . If the number 3 or 5 turns up a subject is selected at random from first group, otherwise the subject is selected from 2^{nd} group. If the probability that an engg. subject is selected is k, then the value of 48k is

PB0111

2. A is one of the 6 horses entered for a race, and is to be ridden by one of two jockeys B or C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win; if C rides A, his chance is trebled. If the odds against his winning is $\frac{p}{q}$ (Where p & q are coprime), then the value of pq is

PB0137

3. The entries in a two-by-two determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ are integers that are chosen randomly and independently, and , for each entry, the probability that the entry is odd is p. If the probability that the value of the determinant is even is 1/2, then the value of $\left[\frac{1000}{p}\right]$ (where [.] denotes greatest integer function) is

PB0117

4. A room has three electric lamps. From a collection of 10 electric bulbs of which 6 are good 3 are selected at random & put in the lamps. If the probability that the room is lighted λ. Then the value of 90λ is

PB0119

5. An author writes a good book with a probability of 1/2. If it is good it is published with a probability of 2/3. If it is not, it is published with a probability of 1/4. The probability that he will get atleast one book published if he writes two is k, then the value of 576k is

PB0131

6. A biased coin which comes up heads three times as often as tails is tossed. If it shows heads, a chip is drawn from urn-I which contains 2 white chips and 5 red chips. If the coin comes up tail, a chip is drawn from urn-II which contains 7 white and 4 red chips. Given that a red chip was drawn. If the probability that the coin came up heads is p. Then the value of $\left[\frac{1}{p}\right]$ (where [.] denotes greatest integer function) is

PB0133

7. A pair of fair dice is tossed. Find the probability that the maximum of the two numbers is greater than 4.

PB0112

8. The chance of one event happening is the square of the chance of a 2^{nd} event, but odds against the first are the cubes of the odds against the 2^{nd}, then the chances of 2^{nd} event happening is (assume that both events are neither sure nor impossible).

PB0127

9. Players A and B alternately toss a biased coin, with A going first. A wins if A tosses a Tail before B tosses a Head; otherwise B wins. If the probability of a head is p and the value of p (for which the game is fair to both players) is $\frac{\sqrt{a}-1}{b}$. Then the value of $\frac{a}{b}$ is

PB0116

10. In a batch of 10 articles, 4 articles are defective. 6 articles are taken from the batch for inspection. If more than 2 articles in this batch are defective, the whole batch is rejected Find the probability that the batch will be rejected.

PB0130

EXERCISE (JM)

1. A multiple choice examination has 5 questions. Each question has three alternative answers of which exactly one is correct. The probability that a student will get 4 or more correct answers just by guessing is : **[JEE-MAIN 2013]**

(1) $\frac{17}{3^5}$ (2) $\frac{13}{3^5}$ (3) $\frac{11}{3^5}$ (4) $\frac{10}{3^5}$

PB0159

2. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$ stands for the complement of the event A. Then the events A and B are : **[JEE(Main)-2014]**

(1) mutually exclusive and independent. (2) equally likely but not independent.

(3) independent but not equally likely. (4) independent and equally likely.

PB0160

3. Let two fair six-faced dice A and B be thrown simultaneously. If E_1 is the event that die A shows up four, E_2 is the event that die B shows up two and E_3 is the event that the sum of numbers on both dice is odd, then which of the following statements is **NOT true** ? **[JEE(Main)-2016]**

(1) E_1, E_2 and E_3 are independent. (2) E_1 and E_2 are independent.

(3) E_2 and E_3 are independent. (4) E_1 and E_3 are independent.

PB0161

4. A box contains 15 green and 10 yellow balls. If 10 balls are randomly drawn, one–by–one, with replacement, then the variance of the number of green balls drawn is :- **[JEE(Main)-2017]**

(1) $\frac{6}{25}$ (2) $\frac{12}{5}$ (3) 6 (4) 4

PB0162

5. If two different numbers are taken from the set {0, 1, 2, 3, ..., 10), then the probability that their sum as well as absolute difference are both multiple of 4, is :- **[JEE(Main)-2017]**

(1) $\frac{7}{55}$ (2) $\frac{6}{55}$ (3) $\frac{12}{55}$ (4) $\frac{14}{45}$

PB0163

6. For three events A, B and C, P(Exactly one of A or B occurs) = P(Exactly one of B or C occurs) = P(Exactly one of C or A occurs) = $\frac{1}{4}$ and P(All the three events occur simultaneously) = $\frac{1}{16}$. Then the probability that at least one of the events occurs, is :- **[JEE(Main)-2017]**

(1) $\frac{3}{16}$ (2) $\frac{7}{32}$ (3) $\frac{7}{16}$ (4) $\frac{7}{64}$

PB0164

7. A bag contains 4 red and 6 black balls. A ball is drawn at random from the bag, its colour is observed and this ball along with two additional balls of the same colour are returned to the bag. If now a ball is drawn at random from the bag, then the probability that this drawn ball is red, is: **[JEE(Main)-2018]**

(1) $\frac{2}{5}$ (2) $\frac{1}{5}$ (3) $\frac{3}{4}$ (4) $\frac{3}{10}$

PB0165

8. An urn contains 5 red and 2 green balls. A ball is drawn at random from the urn. If the drawn ball is green, then a red ball is added to the urn and if the drawn ball is red, then a green ball is added to the urn; the original ball is not returned to the urn. Now, a second ball is drawn at random from it. The probability that the second ball is red, is : **[JEE(Main)-2019]**

(1) $\frac{26}{49}$ (2) $\frac{32}{49}$ (3) $\frac{27}{49}$ (4) $\frac{21}{49}$

PB0166

9. An unbiased coin is tossed. If the outcome is a head then a pair of unbiased dice is rolled and the sum of the numbers obtained on them is noted. If the toss of the coin results in tail then a card from a well-shuffled pack of nine cards numbered 1,2,3,...,9 is randomly picked and the number on the card is noted. The probability that the noted number is either 7 or 8 is : **[JEE(Main)-2019]**

(1) $\frac{13}{36}$ (2) $\frac{19}{36}$ (3) $\frac{19}{72}$ (4) $\frac{15}{72}$

PB0167

10. If the probability of hitting a target by a shooter, in any shot, is 1/3, then the minimum number of independent shots at the target required by him so that the probability of hitting the target at least once is greater than $\frac{5}{6}$, is : **[JEE(Main)-2019]**

(1) 6 (2) 5 (3) 4 (4) 3

PB0168

11. Let S = {1, 2, ... , 20}. A subset B of S is said to be "nice", if the sum of the elements of B is 203. Then the probability that a randomly chosen subset of S is "nice" is :- **[JEE(Main)-2019]**

(1) $\frac{6}{2^{20}}$ (2) $\frac{5}{2^{20}}$ (3) $\frac{4}{2^{20}}$ (4) $\frac{7}{2^{20}}$

PB0169

12. In a random experiment, a fair die is rolled until two fours are obtained in succession. The probability that the experiment will end in the fifth throw of the die is equal to : **[JEE(Main)-2019]**

(1) $\frac{150}{6^5}$ (2) $\frac{175}{6^5}$ (3) $\frac{200}{6^5}$ (4) $\frac{225}{6^5}$

PB0170

13. In a game, a man wins Rs. 100 if he gets 5 of 6 on a throw of a fair die and loses Rs. 50 for getting any other number on the die. If he decides to throw the die either till he gets a five or a six or to a maximum of three throws, then his expected gain/loss (in rupees) is : **[JEE(Main)-2019]**

(1) $\frac{400}{3}$ gain (2) $\frac{400}{3}$ loss (3) 0 (4) $\frac{400}{9}$ loss

PB0171

14. Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is : **[JEE(Main)-2019]**

(1) $\frac{1}{11}$ (2) $\frac{1}{17}$ (3) $\frac{1}{10}$ (4) $\frac{1}{12}$

PB0172

15. If three of the six vertices of a regular hexagon are chosen at random, then the probability that the triangle formed with these chosen vertices is equilateral is : **[JEE(Main)-2019]**

(1) $\frac{3}{10}$ (2) $\frac{1}{10}$ (3) $\frac{3}{20}$ (4) $\frac{1}{5}$

PB0173

16. In a workshop, there are five machines and the probability of any one of them to be out of service on a day is $\frac{1}{4}$. If the probability that at most two machines will be out of service on the same day is $\left(\frac{3}{4}\right)^3 k$, then k is equal to : **[JEE(Main)-2020]**

(1) $\frac{17}{2}$ (2) 4 (3) $\frac{17}{8}$ (4) $\frac{17}{4}$

PB0174

17. If 10 different balls are to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is : **[JEE(Main)-2020]**

(1) $\frac{945}{2^{11}}$ (2) $\frac{965}{2^{11}}$ (3) $\frac{945}{2^{10}}$ (4) $\frac{965}{2^{10}}$

PB0177

18. In a box, there are 20 cards, out of which 10 are lebelled as A and the remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained. The probability that the second A-card appears before the third B-card is : **[JEE(Main)-2020]**

(1) $\frac{11}{16}$ (2) $\frac{13}{16}$ (3) $\frac{9}{16}$ (4) $\frac{15}{16}$

PB0178

19. Four dice are thrown simultaneously and the numbers shown on these dice are recorded in 2×2 matrices. The probability that such formed matrices have all different entries and are non-singular, is : **[JEE(Main)-2021]**

(1) $\frac{45}{162}$ (2) $\frac{23}{81}$ (3) $\frac{22}{81}$ (4) $\frac{43}{162}$ **PB0224**

20. The probability that a randomly selected 2-digit number belongs to the set $\{n \in N : (2^n - 2)$ is a multiple of $3\}$ is equal to **[JEE(Main)-2021]**

(1) $\frac{1}{6}$ (2) $\frac{2}{3}$ (3) $\frac{1}{2}$ (4) $\frac{1}{3}$ **PB0225**

21. A bag contains 4 white and 6 black balls. Three balls are drawn at random from the bag. Let X be the number of white balls, among the drawn balls. If σ^2 is the variance of X, then $100\sigma^2$ is equal to **[JEE(Main)-2022]**

PB0226

22. The probability that a relation R from $\{x,y\}$ to $\{x,y\}$ is both symmetric and transitive, is equal to: **[JEE(Main)-2022]**

(A) $\frac{5}{16}$ (B) $\frac{9}{16}$ (C) $\frac{11}{16}$ (D) $\frac{13}{16}$ **PB0227**

EXERCISE (JA)

1. Four persons independently solve a certain problem correctly with probabilities $\frac{1}{2},\frac{3}{4},\frac{1}{4},\frac{1}{8}$. Then the probability that the problem is solved correctly by at least one of them is

[JEE(Advanced) 2013, 2M]

(A) $\frac{235}{256}$ (B) $\frac{21}{256}$ (C) $\frac{3}{256}$ (D) $\frac{253}{256}$

PB0186

2. Of the three independent events E_1, E_2 and E_3, the probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is γ. Let the probability p that none of events E_1, E_2 or E_3 occurs satisfy the equations $(\alpha - 2\beta)\,p = \alpha\beta$ and $(\beta - 3\gamma)\,p = 2\beta\gamma$. All the given probabilities are assumed of lie in the interval (0,1). Then $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3} =$ **[JEE-Advanced 2013, 4, (–1)]**

PB0187

Paragraph for Question 3 and 4

A box B_1 contains 1 white ball, 3 red balls and 2 black balls. Another box B_2 contains 2 white balls, 3 red balls and 4 black balls. A third box B_3 contains 3 white balls, 4 red balls and 5 black balls.

3. If 2 balls are drawn (without replacement) from a randomly selected box and one of the balls is white and the other ball is red, the probability that these 2 balls are drawn from box B_2 is

(A) $\frac{116}{181}$ (B) $\frac{126}{181}$ (C) $\frac{65}{181}$ (D) $\frac{55}{181}$

[JEE(Advanced) 2013, 3, (–1)]

PB0188

4. If 1 ball is drawn from each of the boxes B_1, B_2 and B_3, the probability that all 3 drawn balls are of the same colour is

(A) $\frac{82}{648}$ (B) $\frac{90}{648}$ (C) $\frac{558}{648}$ (D) $\frac{566}{648}$

[JEE(Advanced) 2013, 3, (–1)]

PB0188

5. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is - **[JEE(Advanced)-2014, 3(–1)]**

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

PB0189

Paragraph For Questions 6 and 7

Box 1 contains three cards bearing numbers, 1,2,3 ; box 2 contains five cards bearing numbers 1,2,3,4,5; and box 3 contains seven cards bearing numbers 1,2,3,4,5,6,7. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the i^{th} box, i = 1,2,3.

6. The probability that $x_1 + x_2 + x_3$ is odd, is -

(A) $\frac{29}{105}$ (B) $\frac{53}{105}$ (C) $\frac{57}{105}$ (D) $\frac{1}{2}$

[JEE(Advanced)-2014, 3(–1)]

PB0190

7. The probability that x_1,x_2,x_3 are in an arithmetic progression, is -

(A) $\frac{9}{105}$ (B) $\frac{10}{105}$ (C) $\frac{11}{105}$ (D) $\frac{7}{105}$

[JEE(Advanced)-2014, 3(–1)]

PB0190

8. The minimum number of times a fair coin needs to be tossed, so that the probability of getting at least two heads is at least 0.96, is **[JEE 2015, 4M, –0M]**

PB0191

Paragraph For Questions 9 and 10

Let n_1 and n_2 be the number of red and black balls respectively, in box I. Let n_3 and n_4 be the number of red and black balls, respectively, in box II.

9. One of the two boxes, box I and box II, was selected at random and a ball was drawn randomly out of this box. The ball was found to be red. If the probability that this red ball was drawn from box II is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1, n_2, n_3 and n_4 is(are)

[JEE 2015, 4M, –0M]

(A) $n_1 = 3, n_2 = 3, n_3 = 5, n_4 = 15$
(B) $n_1 = 3, n_2 = 6, n_3 = 10, n_4 = 50$
(C) $n_1 = 8, n_2 = 6, n_3 = 5, n_4 = 20$
(D) $n_1 = 6, n_2 = 12, n_3 = 5, n_4 = 20$

PB0192

10. A ball is drawn at random from box I and transferred to box II. If the probability of drawing a red ball from box I, after this transfer, is $\frac{1}{3}$, then the correct option(s) with the possible values of n_1 and n_2 is(are) **[JEE 2015, 4M, –0M]**

(A) $n_1 = 4$ and $n_2 = 6$
(B) $n_1 = 2$ and $n_2 = 3$
(C) $n_1 = 10$ and $n_2 = 20$
(D) $n_1 = 3$ and $n_2 = 6$

PB0192

11. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out to be defective. It is known that

P(computer turns out to be defective given that is produced in plant T_1)

= 10P(computer turns out to be defective given that it is produced in plant T_2)

where P(E) denotes the probability of an event E. A computer produces in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is

[JEE(Advanced)-2016, 3(–1)]

(A) $\frac{36}{73}$ (B) $\frac{47}{79}$ (C) $\frac{78}{93}$ (D) $\frac{75}{83}$ **PB0193**

Paragraph For Questions 12 and 13

Football teams T_1 and T_2 have to play two games against each other. It is assumed that the outcomes of the two games are independent. The probabilities of T_1 winning, drawing and losing a game against T_2 are $\frac{1}{2}, \frac{1}{6}$ and $\frac{1}{3}$, respectively. Each team gets 3 points for a win, 1 point for a draw and 0 point for a loss in a game. Let X and Y denote the total points scored by teams T_1 and T_2, respectively, after two games

12. P(X > Y) is- **[JEE(Advanced)-2016, 3(0)]**

(A) $\frac{1}{4}$ (B) $\frac{5}{12}$ (C) $\frac{1}{2}$ (D) $\frac{7}{12}$ **PB0194**

13. P(X = Y) is- **[JEE(Advanced)-2016, 3(0)]**

(A) $\frac{11}{36}$ (B) $\frac{1}{3}$ (C) $\frac{13}{36}$ (D) $\frac{1}{2}$ **PB0194**

14. Let X and Y be two events such that $P(X)=\frac{1}{3}, P(X|Y)=\frac{1}{2}$ and $P(Y|X)=\frac{2}{5}$. Then

[JEE(Advanced)-2017, 4(–2)]

(A) $P(X'|Y)=\frac{1}{2}$ (B) $P(X\cap Y)=\frac{1}{5}$

(C) $P(X\cup Y)=\frac{2}{5}$ (D) $P(Y)=\frac{4}{15}$ **PB0195**

15. Three randomly chosen nonnegative integers x, y and z are found to satisfy the equation $x + y + z = 10$. Then the probability that z is even, is **[JEE(Advanced)-2017, 3(–1)]**

(A) $\frac{36}{55}$ (B) $\frac{6}{11}$ (C) $\frac{5}{11}$ (D) $\frac{1}{2}$ **PB0196**

Paragraph For Questions 16 and 17

There are five students S_1, S_2, S_3, S_4 and S_5 in a music class and for them there are five seats R_1, R_2, R_3, R_4 and R_5 arranged in a row, where initially the seat R_i is allotted to the student S_i, $i = 1, 2, 3, 4, 5$. But, on the examination day, the five students are randomly allotted the five seats.

(*There are two questions based on Paragraph "A". the question given below is one of them*)

16. The probability that, on the examination day, the student S_1 gets the previously allotted seat R_1 and **NONE** of the remaining students gets the seat previously allotted to him/her is -

[JEE(Advanced)-2018, 3(–1)]

(A) $\frac{3}{40}$ (B) $\frac{1}{8}$ (C) $\frac{7}{40}$ (D) $\frac{1}{5}$ **PB0197**

17. For $i = 1, 2, 3, 4$, let T_i denote the event that the students S_i and S_{i+1} do **NOT** sit adjacent to each other on the day of the examination. Then the probability of the event $T_1 \cap T_2 \cap T_3 \cap T_4$ is-

[JEE(Advanced)-2018, 3(–1)]

(A) $\frac{1}{15}$ (B) $\frac{1}{10}$ (C) $\frac{7}{60}$ (D) $\frac{1}{5}$ **PB0197**

18. There are three bags B_1, B_2 and B_3. The bag B_1 contains 5 red and 5 green balls, B_2 contains 3 red and 5 green balls, and B_3 contains 5 red and 3 green balls, Bags B_1, B_2 and B_3 have probabilities $\frac{3}{10}, \frac{3}{10}$ and $\frac{4}{10}$ respectively of being chosen. A bag is selected at random and a ball is chosen at random from the bag. Then which of the following options is/are correct ?

[JEE(Advanced)-2019, 4(–1)]

(1) Probability that the selected bag is B_3 and the chosen ball is green equals $\frac{3}{10}$

(2) Probability that the chosen ball is green equals $\frac{39}{80}$

(3) Probability that the chosen ball is green, given that the selected bag is B_3, equals $\frac{3}{8}$

(4) Probability that the selected bag is B_3, given that the chosen balls is green, equals $\frac{5}{13}$

PB0198

19. Let S be the sample space of all 3×3 matrices with entries from the set $\{0, 1\}$. Let the events E_1 and E_2 be given by

$$E_1 = \{A \in S : \det A = 0\} \text{ and}$$
$$E_2 = \{A \in S : \text{sum of entries of } A \text{ is } 7\}.$$

If a matrix is chosen at random from S, then the conditional probability $P(E_1|E_2)$ equals ____

[JEE(Advanced)-2019, 3(0)]

PB0199

20. Let $|X|$ denote the number of elements in set X. Let $S = \{1,2,3,4,5,6\}$ be a sample space, where each element is equally likely to occur. If A and B are independent events associated with S, then the number of ordered pairs (A,B) such that $1 \leq |B| < |A|$, equals **[JEE(Advanced)-2019, 3(0)]**

PB0200

21. Let C_1 and C_2 be two biased coins such that the probabilities of getting head in a single toss are $\frac{2}{3}$ and $\frac{1}{3}$, respectively. Suppose α is the number of heads that appear when C_1 is tossed twice, independently, and suppose β is the number of heads that appear when C_2 is tossed twice, independently, Then the probability that the roots of the quadratic polynomial $x^2 - \alpha x + \beta$ are real and equal, is

[JEE(Advanced)-2020]

(A) $\frac{40}{81}$ (B) $\frac{20}{81}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$ **PB0206**

22. The probability that a missile hits a target successfully is 0.75. In order to destroy the target completely, at least three successful hits are required. Then the minimum number of missiles that have to be fired so that the probability of completely destroying the target is NOT less than 0.95, is _____

[JEE(Advanced)-2020]

PB0207

23. Two fair dice, each with faces numbered 1,2,3,4,5 and 6, are rolled together and the sum of the numbers on the faces is observed. This process is repeated till the sum is either a prime number or a perfect square. Suppose the sum turns out to be a perfect square before it turns out to be a prime number. If p is the probability that this perfect square is an odd number, then the value of 14p is _____

[JEE(Advanced)-2020]

PB0208

24. Consider three sets $E_1 = \{1, 2, 3\}$, $F_1 = \{1, 3, 4\}$ and $G_1 = \{2, 3, 4, 5\}$. Two elements are chosen at random, without replacement, from the set E_1, and let S_1 denote the set of these chosen elements. Let $E_2 = E_1 - S_1$ and $F_2 = F_1 \cup S_1$. Now two elements are chosen at random, without replacement, from the set F_2 and let S_2 denote the set of these chosen elements. Let $G_2 = G_1 \cup S_2$. Finally, two elements are chosen at random, without replacement, from the set G_2 and let S_3 denote the set of these chosen elements. Let $E_3 = E_2 \cup S_3$. Given that $E_1 = E_3$, let p be the conditional probability of the event $S_1 = \{1, 2\}$. Then the value of p is **[JEE(Advanced)-2021]**

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{1}{2}$ (D) $\frac{2}{5}$ **PB0209**

Question Stem for Question Nos. 25 and 26

Question Stem

Three numbers are chosen at random, one after another with replacement, from the set S = {1, 2, 3, ..., 100}. Let p_1 be the probability that the maximum of chosen numbers is at least 81 and p_2 be the probability that the minimum of chosen numbers is at most 40.

[JEE(Advanced)-2021]

25. The value of $\frac{625}{4}p_1$ is __________. **PB0210**

26. The value of $\frac{125}{4}p_2$ is __________. **PB0211**

27. Let E,F and G be three events having probabilities $P(E)=\frac{1}{8}, P(F)=\frac{1}{6}$ and $P(G)=\frac{1}{4}$ and let $P(E\cap F\cap G)=\frac{1}{10}$. For any event H, if H^C denotes its complement, then which of the following statements is(are) **TRUE** ? **[JEE(Advanced)-2021]**

(A) $P(E\cap F\cap G^C)\le\frac{1}{40}$

(B) $P(E^C\cap F\cap G)\le\frac{1}{15}$

(C) $P(E\cup F\cup G)\le\frac{13}{24}$

(D) $P(E^C\cap F^C\cap G^C)\le\frac{5}{12}$ **PB0212**

28. A number is chosen at random from the set {1, 2, 3, ... , 2000}. Let p be the probability that the chosen number is a multiple of 3 or a multiple of 7. Then the value of 500p is ____.

[JEE(Advanced)-2021]

PB0213

29. In a study about a pandemic, data of 900 persons was collected. It was found that

190 persons had symptom of fever, **PB0228**

220 persons had symptom of cough,

220 persons had symptom of breathing problem,

330 persons had symptom of fever or cough or both,

350 persons had symptom of cough or breathing problem or both,

340 persons had symptom of fever or breathing problem or both,

30 persons had all three symptoms (fever, cough and breathing problem).

If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is _____________.

[JEE(Advanced)-2022]

30. Two players, P_1 and P_2, play a game against each other. In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers. Let x and y denote the readings on the die rolled by P_1 and P_2, respectively. If $x > y$, then P_1 scores 5 points and P_2 scores 0 point. If $x = y$, then each player scores 2 points. If $x < y$, then P_1 scores 0 point and P_2 scores 5 points. Let X_i and Y_i be the total scores of P_1 and P_2, respectively, after playing the ith round.

[JEE(Advanced)-2022]

PB0229

List-I		**List-II**	
(I)	Probability of $(X_2 \geq Y_2)$ is	(P)	$\frac{3}{8}$
(II)	Probability of $(X_2 > Y_2)$ is	(Q)	$\frac{11}{16}$
(III)	Probability of $(X_3 = Y_3)$ is	(R)	$\frac{5}{16}$
(IV)	Probability of $(X_3 > Y_3)$ is	(S)	$\frac{355}{864}$
		(T)	$\frac{77}{432}$

The correct option is:

(A) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)

(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (T)

(C) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (S)

(D) (I) $\rightarrow$ (P); (II) $\rightarrow$ (R); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (T)

31. Suppose that **PB0230**

Box-I contains 8 red, 3 blue and 5 green balls,

Box-II contains 24 red, 9 blue and 15 green balls,

Box-III contains 1 blue, 12 green and 3 yellow balls,

Box-IV contains 10 green, 16 orange and 6 white balls.

A ball is chosen randomly from Box-I ; call this ball b. If b is red then a ball is chosen randomly from Box-II, if b is blue then a ball is chosen randomly from Box-III, and if b is green then a ball is chosen randomly from Box-IV. The conditional probability of the event 'one of the chosen balls is white' given that the event 'at least one of the chosen balls is green' has happened, is equal to

(A) $\frac{15}{256}$ (B) $\frac{3}{16}$ (C) $\frac{5}{52}$ (D) $\frac{1}{8}$

[JEE(Advanced)-2022]

32. Let $X=\left\{(x,y)\in\mathbb{Z}\times\mathbb{Z}:\frac{x^2}{8}+\frac{y^2}{20}<1 \text{ and } y^2<5x\right\}$. Three distinct points P, Q and R are randomly chosen from X. Then the probability that P, Q and R form a triangle whose area is a positive integer, is **[JEE(Advanced)-2023]**

(A) $\frac{71}{220}$ (B) $\frac{73}{220}$ (C) $\frac{79}{220}$ (D) $\frac{83}{220}$ **PB0231**

33. Consider an experiment of tossing a coin repeatedly until the outcomes of two consecutive tosses are same. If the probability of a random toss resulting in head is $\frac{1}{3}$, then the probability that the experiment stops with head is. **[JEE(Advanced)-2023]**

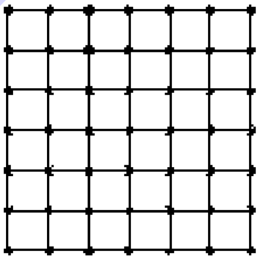
(A) $\frac{1}{3}$ (B) $\frac{5}{21}$ (C) $\frac{4}{21}$ (D) $\frac{2}{7}$ **PB0232**

34. Let X be the set of all five digit numbers formed using 1,2,2,2,4,4,0. For example, 22240 is in X while 02244 and 44422 are not in X. Suppose that each element of X has an equal chance of being chosen. Let p be the conditional probability that an element chosen at random is a multiple of 20 given that it is a multiple of 5. Then the value of 38p is equal to **[JEE(Advanced)-2023]**

PB0233

PARAGRAPH "II"

Consider the 6 × 6 square in the figure. Let $A_1, A_2, \ldots, A_{49}$ be the points of intersections (dots in the picture) in some order. We say that A_i and A_j are friends if they are adjacent along a row or along a column. Assume that each point A_i has an equal chance of being chosen.



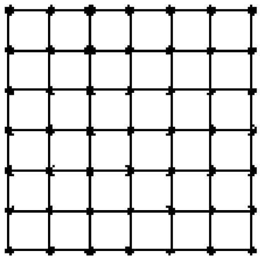
(There are two questions based on PARAGRAPH "II", the question given below is one of them)

35. Let p_i be the probability that a randomly chosen point has i many friends, i = 0,1,2,3,4. Let X be a random variable such that for i = 0,1,2,3,4, the probability $P(X = i) = p_i$. Then the value of 7E(X) is **[JEE(Advanced)-2023]**

PB0234

PARAGRAPH "II"

Consider the 6×6 square in the figure. Let $A_1, A_2, \ldots, A_{49}$ be the points of intersections (dots in the picture) in some order. We say that A_i and A_j are friends if they are adjacent along a row or along a column. Assume that each point A_i has an equal chance of being chosen.



(There are two questions based on PARAGRAPH "II", the question given below is one of them)

36. Two distinct points are chosen randomly out of the points $A_1, A_2, \ldots, A_{49}$. Let p be the probability that they are friends. Then the value of 7p is **[JEE(Advanced)-2023]**

PB0235

ANSWER KEY

Do yourself - 1

1. $\{B_1R_1, B_2R_1, B_3R_1, B_1R_2, B_2R_2, B_3R_2, B_1B_2, B_2B_3, B_1B_3, R_1R_2\}$
2. $\{M_1W_1, M_2W_1, M_1W_2, M_2W_2, M_1W_3, M_2W_3\}$
3. {H1, H2, H3, H4, H5, H6, TH,TT}
4. A

Do yourself - 2

1. $\frac{3}{4}$ **2.** $\frac{10}{21}$ **3.** 1/10 **4.** $\frac{(13)^5-(12)^5}{(13)^5}$ **5.** 23/168

6. (i) 7/13, (ii) 1/2, (iii) 2/13, (iv) 2/13, (v) 1/2, (vi) 9/13

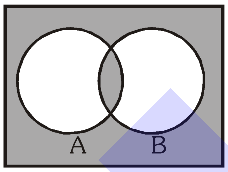
7. 1/56 **8.** 5 : 1 **9.** 4/21 **10.** **(a)** 2/3, **(b)** 1/2

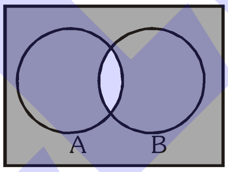
11. B **12.** B **13.** C **14.** B **15.** B

16. 3/4, 1/4; 15/16 **17.** (i) 1/36, (ii) 5/108, (iii) 53/54

18. 11/20 **19.** 3/5 **20.** 2/7 **21.** 12/25

Do yourself - 3

1. **(a)**


(b)


2. A,B,D **3.** 11/15

4. (a) $\frac{19}{25}$ (b) $\frac{13}{25}$ **5.** C **6.** A

7. (i) 0.18, (ii) 0.12, (iii) 0.42, (iv) 0.28, (v) 0.72 **8.** (i) 0.6, (ii) 0.5, (iii) 0.25

9. A **10.** (a) 1/18, (b) 43/90, (c) 5/18, (d) NO **11.** 319/420

12. 17/105

Do yourself - 4

1. A **2.** 1/5525 **3.** C **4.** D **5.** B

6. A **7.** C **8.** B

Do yourself - 5

1. A, B **2.** 1/36 **3.** C **4.** C **5.** A

6. B **7.** C **8.** B

Do yourself - 6

1. 2/21 2. $\frac{1}{n+2}$ 3. C 4. A 5. C
6. C 7. A 8. D

Do yourself - 7

1. **(a)** 0.72 **(b)** 0.23

Do yourself - 8

1. $\frac{496}{729}$ 2. ${}^{7}C_{3}\left(\frac{1}{6}\right)^{3}\left(\frac{5}{6}\right)^{4}$ 3. D

Do yourself - 9

1. 2/3 2. 9/25 3. B 4. A 5. C
6. C 7. C 8. A 9. C

Do yourself - 10

1. A 2. B 3. B 4. A 5. B 6. C

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	C	A	D	B	C	A	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	A	A	C	A	C	C	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	B	B	D	D	4	1	1	3

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C,D	A,B,C,D	A,B,D	A,C,D	B,C,D	A,D	B,C,D	A,D	A,B,C,D	A,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,D	B,D	A,B	A,D	C,D	A,B	A,B,C	B,D	A,B,C,D	A,D
Que.	21	22	23	24	25					
Ans.	A,B,C,D	A,B	A,B	A,C,D	A,B					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	C	B	C	C	0.36	0.25	A,B,D	B,C,D	A
Q.10	A	B	C						
	Q	P	S						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	26	65	1414	87	407	1.00	0.55	0.33	2.50	0.45

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	1	2	2	3	1	2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	3	1	2	3	NTA Ans. (3); Correct Ans. is $\frac{17 \times 945}{2^{15}}$	1	4	3
Que.	21	22								
Ans.	56	A								

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	6	D	A	A	B	C	8	A,B	C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	C	A,D	B	A	C	2,3	0.50	422.00
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	6	8.00	A	76.25	24.50	A,B,C	214	0.80	A
Que.	31	32	33	34	35	36				
Ans.	C	B	B	31	24.00	0.50				